Factoring Polynomials

Concept

Factoring reverses expansion: write a sum as a product.

First step

Take out the greatest common factor.

Trinomial

For $x^2 + bx + c$, find $p + q = b$ and $pq = c$.

Square difference

$a^2 - b^2 = (a - b)(a + b)$.

Finish

Factor completely; expand to check.

Example 1

Factor $x^2 - x - 20$.

Solution

1. Find numbers with product -20 and sum -1: -5 and 4.

2. Result: $(x - 5)(x + 4)$.

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Example 2

Factor $3x^3 - 27x$ completely.

Solution

1. Take out $3x$: $3x(x^2 - 9)$.

2. Factor the square difference: $3x(x - 3)(x + 3)$.

Factoring Polynomials

Practice | your turn

Show your steps. Include units and restrictions where needed.

1. Factor $8x^2 - 12x$.

2. Factor $x^2 + x - 30$.

3. Factor $x^2 - 10x + 25$.

4. Factor $4x^2 - 49$.

5. Explain why $x^2 + 9$ is not $(x + 3)(x - 3)$.

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Practice answers | check after trying

1. $4x(2x - 3)$.

2. $(x + 6)(x - 5)$.

3. $(x - 5)^2$.

4. $(2x - 7)(2x + 7)$.

5. The proposed product expands to $x^2 - 9$; its constant has the wrong sign.